KIT-PROJECTS · STARTER · ANNOTATED SKETCH

Modulino starter sketch

The smallest useful Modulino program: press button A to light a pixel green, press B to turn it off. Every project in this series has the same shape - include the library, create the modules, start them, then use them in `loop()`.

BEFORE YOU START

A. Install the library: in the Arduino IDE choose **Sketch → Include Library → Manage Libraries...**, search for `Modulino` and install **Arduino_Modulino**. If asked to install its dependencies, choose **Install all**.

B. Choose the board: **Tools → Board → Arduino UNO R4 WiFi**, and select its USB port.

C. Plug the Modulino Buttons and Modulino Pixels into the Uno R4's **Qwiic** connector, chained one after the other. No other wiring.

THE SKETCH, LINE BY LINE

1

```
#include <Arduino_Modulino.h>
```

Pulls in the Modulino library. This one line gives you a class for every module (`ModulinoButtons`, `ModulinoKnob`, ...) and the `Modulino` object that starts the bus.

2

```
ModulinoButtons buttons;
ModulinoPixels pixels;
```

One object per module you use. Name them anything you like. There is only one of each type in the kit, so the library finds each module on the Qwiic bus by itself - no addresses needed.

3

```
Serial.begin(115200);
```

Lets the sketch print messages. Open **Tools → Serial Monitor** (set to 115200) to read them.

4

```
Modulino.begin();
```

Starts the Qwiic bus. **Always call this first**, before any module's `begin()`.

5

```
if (!buttons.begin())
  Serial.println("Buttons not found");
```

Each module has its own `begin()`. It returns `false` if the module isn't found, which is a quick way to spot a loose Qwiic cable.

6

```
if (buttons.update()) {
```

`update()` asks the module for its latest state and returns `true` **only when something changed**. Call it once every time through `loop()`.

7

```
if (buttons.isPressed('A')) {
```

Which button? `'A'`, `'B'` or `'C'`. Other modules have their own calls - see the table on the next page.

8

```
pixels.set(0, GREEN, 40);
pixels.show();
```

`set(pixel, color, brightness)`: pixel 0-7, a color name (or `ModulinoColor(r, g, b)`), brightness 0-100. **Nothing lights up until you call** `show()`. **To turn a pixel off, use** `pixels.clear(0)` and call `show()` again.

THE SEVEN MODULES AT A GLANCE

Module	Class name	Common calls
Buttons	`ModulinoButtons`	`update()`, `isPressed('A')`, `setLeds(a, b, c)`
Knob	`ModulinoKnob`	`getDirection()` (+1 / -1 / 0), `isPressed()`, `get()`
Buzzer	`ModulinoBuzzer`	`tone(frequency, ms)`, `noTone()`
Pixels	`ModulinoPixels`	`set(i, color, brightness)`, `show()`, `clear(i)`
Distance	`ModulinoDistance`	`available()`, `get()` (millimeters)
Movement	`ModulinoMovement`	`update()`, `getRoll()` / `getPitch()` / `getYaw()`, `getX()` / `getY()` / `getZ()`
Thermo	`ModulinoThermo`	`getTemperature()`, `getHumidity()`

THINGS THAT TRIP PEOPLE UP

- **Nothing happens at all?** Check the Qwiic cables are pushed fully in at both ends. A loose connector looks exactly like a missing module, and `begin()` returns `false`.
- **A change is only reported once.** Call `update()` once per loop and share the result - calling it again straight away always says “nothing changed”.
- **Avoid long delays.** A long `delay()` freezes the sketch, so it misses button presses and sensor readings. Use `millis()` timing instead (project 2 shows how).
- **The Knob’s raw position is arbitrary.** `get()` starts at a random-looking number and counts very finely. Use `getDirection()` instead.
- **No reading when nothing is in range.** The Distance sensor has nothing to report then, so check `available()` before calling `get()`.

WHERE NEXT

Work through projects 1-6 in order; each one adds a new module or idea. To clear the board between projects, flash `00-hello-world`.

THE COMPLETE SKETCH

Also in `starter-sketch/starter-sketch.ino` - this page is built from that file.

```
/*
 * Plug and Make Kit - Starter sketch: using the Modulino library
 *
 * The smallest useful Modulino sketch. Press button A on the Modulino
 * Buttons module to light the first pixel on the Modulino Pixels module
 * green; press button B to turn it off again.
 *
 * Wiring: plug the Modulino Buttons and Modulino Pixels into the Uno R4's
 * Qwiic connector (chain them together). Nothing else is needed.
 *
 * Requires the "Arduino_Modulino" library (Library Manager).
 */

#include <Arduino_Modulino.h>

// One object per module. There is only one of each type in the kit, so the
// library finds each module on the Qwiic bus by itself - no addresses needed.
ModulinoButtons buttons;
ModulinoPixels pixels;

void setup() {
  Serial.begin(115200);

  Modulino.begin();  // start the Qwiic bus - always do this first

  // Each module's begin() returns false if that module isn't found.
  if (!buttons.begin()) Serial.println("Buttons module not found - check the Qwiic cable");
  if (!pixels.begin())  Serial.println("Pixels module not found - check the Qwiic cable");
}

void loop() {
  // update() returns true only when a button has just been pressed or released.
  if (buttons.update()) {
    if (buttons.isPressed('A')) {
      pixels.set(0, GREEN, 40);  // pixel number, color, brightness (0-100)
      pixels.show();             // nothing changes until you call show()
    } else if (buttons.isPressed('B')) {
      pixels.clear(0);           // turn pixel 0 off
      pixels.show();
    }
  }
}
```